2025110840 - 高思羽 - 机械04

西南交通大学
Southwest Jiaotong University

1. 创建一个 k值全为 10 含有0的一维数组.让角标为奇数的元素等于1.

```
import numpy as np
a = np.zeros(10)
a[4] = 1
print(a).
```

2. 生成一个 4×5 和 5×4 的矩阵.计算他们的乘积.

```
import numpy as np.
rng = np.random.default_rng()
a = rng.integers(1, 11, size=(4,5))
b = rng.integers(1, 11, size=(5,4))
c = a@b.
print(a, "\n", "*", "\n", "b", "\n", "=", "\n", c)
```

3. 函数 $f(x)=(\sin(x+1)/e^x)(x+1)/2$.生成一个数组 $a \in R^{5}$.计算 $b=f(a)$

```
import numpy as np
~~import math~~
a = np.arange(1,6,dtype=float).reshape(1,5)
def f(a):
    x = a
    result = (np.sin(x+1)/np.exp(x) * (x+1)/2)
    return result

b = f(a)
print("a=", a)
print("b=", b)
```

4. 随机生成一个 5行 10列的矩阵，然后每行元素一步法行平均值.

```
import numpy as np
rng = np.random.default_rng()
a = rng.integers(1, 11, size=(5,10)).astype(float)
b = np.zeros(5)
sum = 0
for i in range(5):
    for j in range(10):
        sum += a[i,j]
    b[i] = sum/10
    sum = 0
for in in range(5):
    for j in range(10)
        a[i,j] = b[i]
print(a)
```

2025110840 - 高思羽 - 机械04

西南交通大学
Southwest Jiaotong University

1. 用蓝线绘制函数 $f(x)=x^2-2x+3$ 的图象，并在坐标(1,2)处标个红点。

```
import matplotlib.pyplot as plt
import numpy as np
x = np.linspace(-10,10,2000)
y = x**2 - 2*x +3
plt.plot(x,y, color = "blue", label = "fx) - x^2 -2x+3")
plt.scatter(1,2, color = "red", zorder =5)
plt.title("fx) = x^2 -2x+3")
plt.xlabel("x")
plt.ylabel("fx)")
plt.grid = (True)
plt.legend()
plt.show
```

2. 使用 np.linspace() 函数生成 $t\in[0,2\pi]$，给定 $x=16\sin(t)^3$ 和 $y=13\cos(t)-5\cos(2t)-2\cos(3t)-\cos(4t)$，画出 x-y 图象，并将图象加个红色 Heart。

```
import numpy as np
import matplotlib.pyplot as plt
t = np.linspace(0,2*np.pi)
x = 16*(np.sin(t)**3)
y = 13*np.cos(t) -5*np.cos(t*2) -2*np.cos(3*t) -np.cos(4*t)
plt.plot(x,y, color = "pink", label = "Heart")
plt.title("Heart")
plt.xlable("x")
plt.ylabel("y")
plt.grid("True")
plt.legend()
plt.show
```

① 导入库. ② 读取数据. ③ ~~导入库~~ 显示前5行. 后5行.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
%matplotlib inline
titanic = pd.read_csv("titanic.csv")
print("数据集前5行:")
display(titanic.head())
print("后5行:")
display(titanic.tail())
```

④ 查询行的、列为列?

```
print(f"行数 = {row}")
print(f"列数 = {columns}")
```

⑤ 将 PassengerID 设置为索引.

```
titanic.set_index("PassengerID")
```

⑥ 数据中是否存缺失值.

```
print(titanic.isnull().sum())
print(titanic.isnull().values.any())

titanic = titanic.set_index("PassengerID")
```

⑦ 乘客最大年龄和最小年龄是多少.

```
print(titanic["Age"].max())
print(titanic["Age"].min())
```

⑧ 存活为人性比：

```
print(titanic["Survived"].sum())
```

⑨ 男女生存比例合影比较：

```
femal_rate = titanic.loc[titanic["Sex"]=="female", "Survived"].mean()
print(femal_rate).
```

⑩ 按船票价格降序排列.

```
titanic_sorted = titanic.sort_value(by="Fare", ascending=False)
display(titanic_sorted)
```

⑪. 绘制一个展示船票价格的直方图.

```
plt.hist(titanic["Fare"].dropna(), bins=20, edgecolor="black")
plt.xlabel("Fare")
plt.ylabel("Passenger Count")
plt.title("Distribution of Ticket Fares")
~~plt~~
plt.show(),
```

⑫ 绘制乘客年龄与船价格的散点图.

```
plt.scatter(titanic["Age"].titanic["Fare"], alpha=0.5)
plt.xlabel("Age")
plt.ylabel("Fare")
plt.title("Passenger Age vs. Ticket Fare")
plt.show()
```

西南交通大学
Southwest Jiaotong University

1.1 xxx one said (famous_person, message).

famouse_person "岳飞中"
message = "滚滚长江东逝水"
print(famous_person, "once said", "message", "")

1.2. names = ["Bob", "Charlie", "Alice", "David"]

```
for name in names
    print(name)
print("\n", "排序后的姓名:")
names.sort()
for name in names:
    print(name)

names.append("Albert Einstein")
length = len(names)
print("\n", "列表长度=", length).
```

1.3. numbers = list(range(3,31,3))

```
for num in numbers:
    print(num)
```

1.4. usernames = ['admin', 'Bob', 'Charlie', 'Jude', 'David']

```
for username in usernames
    if username == "admin":
        print("~~Hello~~" "Hello", username, "would you like to see a status report?")
    else:
        print("Hello", username, "thank you for logging in again.")
```

1.5.

```
import pandas as pd.
import numpy as np
rng=np.random.default_rng()
L = rng.integers(-10, 10, size =(1,5))
Lup=np.sort(L)
Ldown= np.sort(L, axis = 1)
print(2)
if np.array_equal(L,Lup):
    print("UP")
elif np.array_equal(L, Ldown):
    print("Down")
else:
    print("WRONG")
```

1.6

```
S= input("请输入 IP 地址:")
parts = S.split(".")
flag = len(parts) == 4
for part in parts:
    if(
        not part
        or not all("0"<=ch<="9" for ch in part)
        or (len(part)>1 and part[0]=="0")
        or int(part) > 255
    ):
        flag = False
        break
print("Yes" if flag else "No")
```

2025110840 - 高思翔 - 机械04

西南交通大学
Southwest Jiaotong University

1.7

```
sandwich_orders = ["tuna", "egg", "ham"]
finished_sandwiches = []

while sandwich_orders:
    current_sandwich = sandwich_orders.pop()
    print(f"I made your {current_sandwich} sandwich.")
    finished_sandwich.append(current_sandwich)
print("\nThe following sandwiches were made:")
for sandwich in finished_sandwiches:
    print(sandwich)
```

1.8

```
cities = {
    "東京": {
        "country": "日本",
        "population": "约1600万",
        "fact": "東京是日本的首都。"},
    "巴黎": {
        "country": "法国",
        "population": "约210万。",
        "fact": "巴黎有埃菲尔铁塔闻名。"},
    "开罗": {
        "country": "埃及",
        "population": "约1000万",
        "fact": "开罗附近有吉萨金字塔。"}}
for city, information in cities.items():
    print(f"\n城市: {city}")
    print(f"国家: {information['country']}")
    print(f"人口: {information['population']}")
    print(f"事实: {information['fact']}")
```

1.9

```
t = {
    "year": "2013",
    "month": "9",
    "day": "30",
    "hour": "16",
    "minute": "45",
    "second": "2"
    }
result = (
    f"{t["year"].zfill(4)}-"
    f"{t["month"].zfill(2)}"
    f"{t["day"].zfill(2)}"
    f"{t["hour"].zfill(2)}:"
    f"{t["minute"].zfill(2)}:"
    f"{t["second"].zfill(2)}"
    )

print (result)
```

1.10

```
def favorite_book(title):
    print(f"One of my favorite book is {title}.")

favorite_book("Alice in Wonderland")
```

西南交通大学
Southwest Jiaotong University

2.1.1
```
squares = [number ** 2 for number in range(1, 11)]
print(squares)
```

2.1.2
```
num = [number for number in range(1,21) if number %2 == 0]
print(num)
```

2.1.3
```
matrix = [[row+col for col in range(3)] for row in range(3)]
print(matrix)
```

2.1.4
```
word = ["hello", "python", "good morning"]
capitalized_words = [word.capitalize() for word in words]
print(capitalized_words)
```

2.1.5
```
num = [number for number in range(1,51) if number %3 == 0 or
number %5 == 0]
print(num)
```

2.1.6
```
scores = {"Alice": 15, "Bob": 8, "Charlie": 20}
key_over_10 = [key for key, value in scores.items() if value >10]
print(keys_over_10)
```

2.2.1
```
class Person:
    def __init__(self, name, age):
        self.name = name
        self.age = age
    def introduce(self):
        print(f"大家好，我叫{self.name}，今年{self.age}岁。")
person = Person("小明", 20)
person.introduce()
```

2025 11 08日 - 高思羽 - 机械04

2.2.2

```
class Person:
    def __init__(self, name, age):
        self.name = name
        self.age = age

    def introduce(self):
        print(f"大家好，我叫{self.name}，今年{self.age}岁。")

class Student(Person):
    def __init__(self, name, age, student_id):
        super().__init__(name, age)
        self.student_id = student_id

    def study(self):
        print(f"{self.name}（学号：{self.student_id}）正在学习")

student = Student("小明", 20, "S001")
student.introduce()
student.study
```

2.2.3

```
class Animal:
    def speak(self):
        print("动物发出声音")

class Dog(Animal):
    def speak(self):
        print("汪汪")

class Cat(Animal):
    def speak(self):
        print("喵喵")

dog = Dog()
cat = Cat()
dog.speak()
cat.speak()
```

2025.10.8 - 高迎羽 - 机械 0 4

2.2.4

```
class MathUtils:
    @classmethod
    def add(cls,a,b):
        return a+b
    @staticmethod
    def multiply(a,b):
        return a*b
print(MathUtils.add(3,5))
print(MathUtils.multiply(3,5))
```

2.2.5

```
class Circle:
    def __init__(self,radius):
        self.radius=radius
    @property
    def radius(self)
        return self._radius
    @radius.setter
    def radius(self,value):
        if value<0:
            raise ValueError("半径不能为负数")
        self._radius=value
circle=Circle(5)
print(circle.radius)
circle.radius=3
print(circle.radius)
```

2.2.6

```
class Swimmer:
    def swim(self):
        print("会游泳")
class Flyer
    def fly(self)
        print("会飞")
```

20251108 40 - 高旻羽 - 作業04

```
class FlyingFish(Swimmer, Flyer):
    pass

fish = FlyingFish()
fish.swim()
fish.fly()
```

2.2.7

```
class Point:
    def __init__(self, x, y):
        self.x = x
        self.y = y
    def __add__(self, other):
        return Point(self.x + other.x, self.y + other.y)
    def __str__(self):
        return f"({self.x}, {self.y})"

p1 = Point(2,3)
p2 = Point(4,5)
print(p1 + p2)
```

2.2.8

```
class Logger:
    _instance = None
    def __new__(cls):
        if cls._instance is None:
            cls._instance = super().__new__(cls)
        return cls._instance
    def log(self, message):
        print(f"[LOG] {message}")

logger1 = Logger()
logger2 = Logger()

~~logger1.~~
Logger1.log("開始自動")
~~logger2.log~~
print(logger1 is logger2)
```

2025110840 - 蒲思羽 - 机械0中

西南交通大学
Southwest Jiaotong University

2.2.9

```
from abc import ABC, abstractmethod
from math import pi
class Shape(ABC):
    @abstractmethod
    def area(self):
        pass

class Circle(Shape):
    def __init__(self, radius):
        self.radius = radius
    def area(self):
        return pi * self.radius **2

class Rectangle(Shape):
    def __init__(self, width, height):
        self.width = width
        self.height = height
    def area(self):
        return self.width * self.height

circle = Circle(3)
rectangle = Rectangle(4,5)
print(circle.area())
print(rectangle.area())
```

2.3.1

```
def matrix_multiply(a,b):
    if not a or not b or len(a[0]) != len(b):
        raise ValueError("矩阵维度不匹配，无法相乘")

    rows = len(a)
    cols = len(b[0])
    shared = len(b)

    result = [[0 for _ in range(cols)] for _ in range(rows)]

    for i in range(rows):
        for j in range(cols):
            for k in range(shared):
                result[i][j] += a[i][k] * b[k][j]

    return result

a = [
    [1,2,3],
    [4,5,6]
    ]

b = [
    [7,8],
    [9,10],
    [11,12]
    ]

print(matrix_multiply(a,b))
```